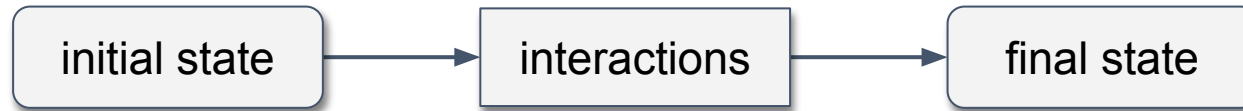


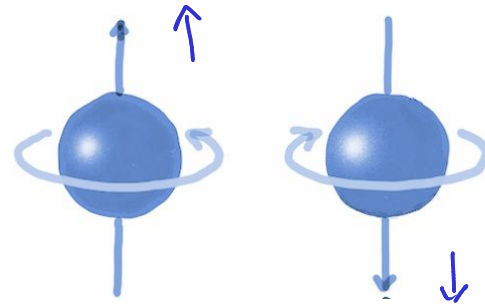
An overview of computation



(I)
 $|0\rangle$

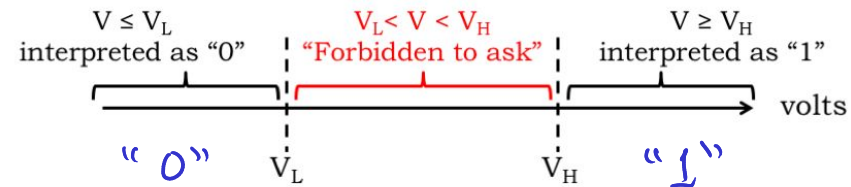
(E)

(M) →



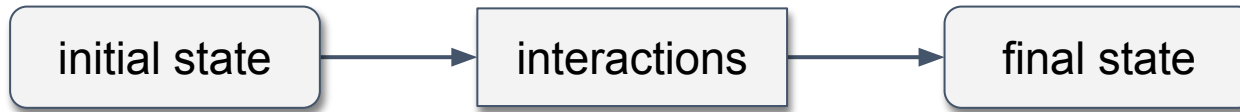
$$|\psi\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle$$

Physical System: Electron (quantum)

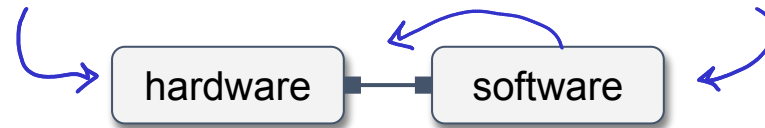


Physical System: Digital Electronic Circuit (classical)

An overview of computation

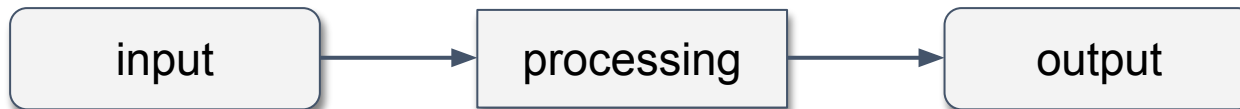


- Information is **encoded** in the state of a physical system
- Computation can be modeled as a sequence of state-changes
 - Physical properties are mapped to mathematical framework

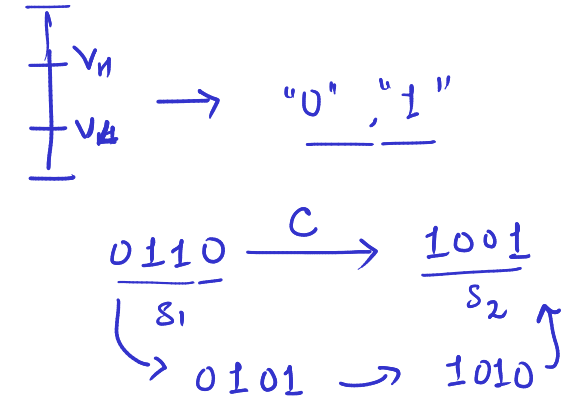


- What mathematical operations can be carried out?
 - only those that can be mapped to physical processes
 - there are physical limits to computation
 - engineering limits
 - fundamental limits

- 'Input, processing, output' is an abstraction of this physical process

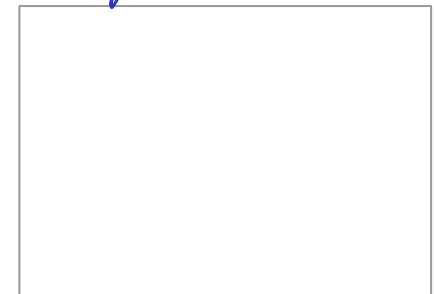


→ Initial → $s_1 s_2 s_3 \dots (s_n)$ → output.

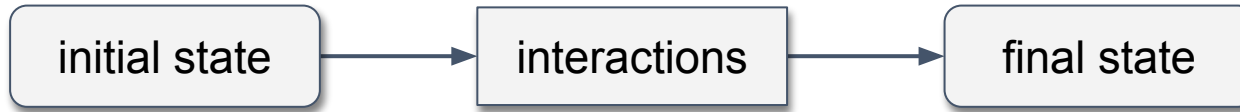


classical
 $\{0,1\}$
 mixing of states
 quantum tunneling ✓

thermal fl. decoherence } quantum state



An overview of computation

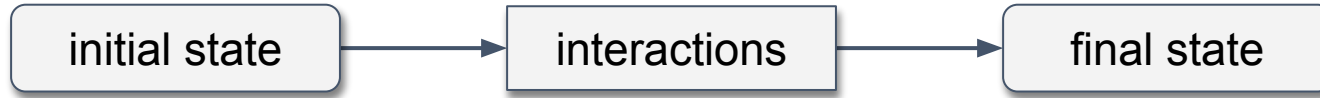


- Availability of different physical interactions
 - many different types of 'computers' can be designed
- ✓ • Optimisation of physical resources
 - required for implementing a given task
 - optimal sequence of state-changes on a given physical system?

"better processors"

- ✓ 1. "Physics and Computation", T. Toffoli, Int. J. Theor. Phys. 21, 165-175 (1982). (I) *} fundamental*
- 2. "Information is Physical", R. Landauer, Phys. Today 44, 23-29, May (1991). (I) *} $\Rightarrow$ IT, entropy*
- 3. "Feynman Lectures on Computation", Richard Feynman, 1996 *}*

An overview of computation



“Models”

- Theory of computation is a subset of the theory of physics
- Information and computation are fundamentally physical
 - Example: Reversibility of computations

“Models”

$U U^\dagger$

$\neg$ (NOT)
 $\wedge$ (AND)

x1	x2	y
0	0	0
0	1	0
1	0	0
1	1	1

$f(x_1, x_2) \rightarrow y$
 “irreversible”
 losing information

Computing
 Thermodynamics
 Information Theory

$x_1 = 1, x_2 = 1$

Exercise:

1. Learn about reversible computing
2. What is a Toffoli gate?

classical - quantum

1. R. Landauer, "Irreversibility and Heat Generation in the Computing Process," in *IBM Journal of Research and Development*, vol. 5, no. 3, pp. 183-191, July 1961, doi: 10.1147/rd.53.0183.
2. "Reversible Computing", Simons Conference Lecture, Cousin Michael P. Frank, 2003